

FROM IDEA TO INSTALLED OBJECT

One continuous experience · A demonstration case, not an operating system

One Continuous Experience

Sarah has lived in her house for eight years. Next to the fireplace is a narrow alcove where nothing fits. Too deep for standard furniture. Not square enough for shelving. The left wall bows inward by half an inch, and the baseboard cuts diagonally across the back. The floor slopes slightly where the foundation settled.

Every attempt ends the same way: measure, guess, trim, and return pieces that “almost” worked. Nothing was wrong with the parts. They were simply correct somewhere else.

One evening she opens the interface. It asks how she wants to capture the space: enter measurements, upload plans, or scan. She uses her phone. The screen traces wall lines, ceiling angle, and floor slope. It pauses to confirm two boundaries she passed too quickly. Then it asks three alignment questions:

- Should components align with the fireplace mantle?
- Should the system compensate for the floor slope?
- Should trim match existing molding?

She selects yes. She has not drawn anything, yet the space is already defined. A guided interface walks her from capture through layout and material choice. She is not working as a drafts person or engineer. The steps that would normally require a designer are handled in the exchange between her choices and the system. She chooses a layout, then adjusts the shelf heights so the upper shelves line up with the fireplace mantle and the bottom shelf is deep enough for the objects she keeps there. She upgrades from the default pine to North Carolina cherry. The price is not a loose estimate. It is built from her selections, the measured space, and available local inventory, and it updates as she changes material or layout.

Those intermediate translation steps are compressed into a single resolved definition. A confirmation appears: **Project Buildable**. All design elements are within defined system parameters. She approves the definition. The fabrication definition is confirmed before anything is cut.

At the lumberyard

The definition is transmitted to a nearby lumberyard. The operator does not interpret drawings; they load the boards listed on the screen. The machine references each piece and begins only the operations the yard’s bounded cell is equipped to perform. Ends are cut to account for the floor slope she scanned but never calculated. Fastener locations are taken from the resolved packet rather than marked on site.

At home, she opens the bundle. The instructions contain no dimensions, only relationships: **A1 connects to A2; B3 inserts into B2**. The lowest component sits level despite the floor slope. The side panels meet the ceiling evenly. She keeps waiting for a piece that requires shaving or forcing, but it never arrives. The final piece slides in without trimming.

The project succeeds because the room was resolved before the saw began. Sarah realizes she hasn’t touched a tape measure. The object was not adapted to the room after the fact; it arrived already adapted.

Workflow: scan → define → select → confirm → cut–mill–drill → label → pickup → install

The fabrication definition exists at the moment of confirmation. Nothing new is designed after that point.

THE MECHANISM: WHY THE RESULT IS RELIABLE

1. Orchestration: resolution before cutting

The result Sarah experienced is produced by a process resolved at confirmation. At that moment the system generates three outputs together: a SKU-bound bill of materials, machine-ready instructions for the operations the local cell will accept, and labeled assembly relationships.

This is not a drawing file. It is a **resolved fabrication definition** that reconciles spatial capture, geometric constraints, and available material. Because geometry and inventory are reconciled in advance, price can be known before cutting, and execution confirms a definition that already exists. After confirmation, no new design interpretation is required. The yard still decides material condition, safety, setup, and whether the job stays in its declared window. Reliability is established before the first cut.

2. The two-patent structure: definition layer + execution layer

The system addresses two independent failure modes: an incomplete fabrication definition, and dimensional drift during execution.

- **Patent 1 (U.S. 9,720,401)** covers the upstream translation: converting intent into a complete fabrication definition tied to real material, with relevant geometric and material constraints resolved before anything moves.
- **Patent 2 (U.S. 10,768,609)** covers bounded execution: operations re-anchor to a shared geometric origin so accuracy does not depend on accumulated field measurement.

The first patent establishes that the workflow can exist; the second establishes that it can be executed repeatably in a supervised environment.

3. The bounded cell: fewer options, so a yard can run it

The cell occupies two familiar shop footprints: a dimensional station matching a radial-arm saw path, and a sheet station matching a vertical panel unit. It is not a general-purpose CNC. Its domain is fit-critical interior assemblies, work that must lock to an existing opening, not open-ended millwork.

Tooling is a small fixed set of saws, drills, routers, and mills. Positions and datums are mapped to the fixture once. When a resolved packet arrives, the machine already knows which tool to call and where it is. No per-job tool changes. No manual datum setting. No programmer setting offsets for each order. After setup, the yard loads the listed stock and starts the job, or refuses it if it falls outside the declared window. The fixed tool geometry and bounded operation library eliminate per-job CAM authoring at the yard.

4. Distributed infrastructure

Because each unit mirrors existing shop footprints, evaluation does not require a new building. The hardware is intended to sit in material-handling paths retailers and institutions already maintain: a lumberyard, a regional shop, or a training lab, without structural retrofit.

A narrow technical boundary with a wide operational question allows the system to be examined in three bounded contexts: as a supervised fabrication training cell within an existing lab footprint; as a limited manufacturability pilot in a regional shop or lumberyard; or as a research platform for constraint-based fabrication workflows. It is offered for structural evaluation, not as a claim that the service is already running.